Formula 1 Data Analysis – Capstone Project Questions

Project Overview

In this capstone, you will use real Formula 1 data to practice relational database design and SQL analytics. You will work with the `formula_1_2025_complete` dataset and write queries to answer analytical questions about drivers, constructors, races, and results.

This project assesses your ability to:

- Design an Entity–Relationship Diagram (ERD) (Optional)
- Create a relational database schema (Optional)
- Write SQL queries using `JOIN`, `GROUP BY`, `ORDER BY`, `CASE`, and subqueries
- Reason critically about multi-table relationships
- Hints are provided sparingly — think critically and reason through each task.
- Analyze real data to answer applied questions

Choose Your Approach

Option A — Full Challenge (Recommended)

If you want the full learning experience:

1. **Ignore** the (full_schema.sql) file.
2. Design **your own ERD** based on the Formula 1 dataset.
3. Create all tables manually (keys, constraints, relationships).
4. Load the dataset into your schema. This is the (load data local infile.sql) file
5. Complete all SQL analysis questions using your own database design.

This option gives the most practice with real database architecture.

Option B — Assisted Path (Use Provided Schema)

If you get stuck or want to focus on analysis:

1. Use the (full_schema.sql) file as provided.
2. Run it to create all tables and relationships automatically.
3. Load the dataset into your schema. This is the (load data local infile.sql) file
4. Begin working directly on the SQL analysis questions.

This option ensures you can complete the project even if database design becomes difficult.

---

1. How many races were held in each decade?

Objective: Count the total number of races grouped by decade.

Hints:

- Consider grouping years into ranges representing decades.
- Think about how to label years that don't fall into standard decades.
- You will need to summarize counts per group.
- **Expected :**

decade	number_of_races
1950s	84
1960s	100
1970s	144
1980s	156
1990s	162
2000s	174
2010s	198
2020s	107

---

2. Identify F1 Powerhouses – Constructors with 100+ Wins

Objective: Find constructors with more than 100 race wins.

Hints:

- Wins are determined by race outcomes.
- Consider how to combine constructor details with race results.
- Aggregate the total wins per constructor and filter for high performers.
- **Expected :**

constructor_name	total_wins
Ferrari	249
McLaren	185
Mercedes	129
Red Bull	122
Williams	114

---

3. Identify Packed F1 Seasons – Years with Over 20 Races

Objective: Determine which seasons had the most races.

Hints:

- Summarize the number of races per year.
- Filter years where the total count exceeds a certain threshold.
- **Expected :**

year	race_count
2016	21
2018	21
2019	21
2021	22
2022	22
2023	22
2024	24

---

4. Identify Drivers with Over 200 Career Points

Objective: List drivers who accumulated more than 200 points in their careers.

Hints:

- Sum points for each driver across all races.
- Focus on drivers who meet the threshold.
- **Expected :**

driverId	total_points
1	4820.50
2	259.00
3	1594.50
4	2329.00
8	1873.00
9	274.00
13	1167.00
14	535.00
15	246.50
17	1047.50
18	1235.00
20	3098.00
21	275.00
22	658.00
23	329.00
30	1566.00
31	307.00
35	235.00
55	241.00
57	420.00
71	360.00
77	385.00

---

5. Find Races Held at High-Altitude Circuits (Above 800 Meters)

Objective: Identify races held on circuits with challenging altitudes.

Hints:

- First, determine which circuits are above a certain altitude.
- Then identify races associated with those circuits.
- Include relevant details for each race.
- **Expected :**

race_name	year
Mexican Grand Prix	1963
Mexican Grand Prix	1964
Mexican Grand Prix	1965
Mexican Grand Prix	1966
Mexican Grand Prix	1967
South African Grand Prix	1967
Mexican Grand Prix	1968
South African Grand Prix	1968
Mexican Grand Prix	1969
South African Grand Prix	1969
Mexican Grand Prix	1970
South African Grand Prix	1970
South African Grand Prix	1971
South African Grand Prix	1972
South African Grand Prix	1973
South African Grand Prix	1974
South African Grand Prix	1975
South African Grand Prix	1976

---

6. Categorize Driver Performance Tiers Based on Career Points

Objective: Rank drivers by their career points.

Hints:

- Calculate total points per driver.
- Think about how to divide drivers into categories (e.g., high, medium, low).
- Consider how SQL can classify values into ranges.
- **Expected :**

driverId	total_points	performance_category
1	4820.50	Elite
2	259.00	Good
3	1594.50	Elite
4	2329.00	Elite
5	105.00	Good
6	9.00	Average
7	6.00	Average
8	1873.00	Elite
9	274.00	Good
10	51.00	Average
11	44.00	Average
12	19.00	Average
13	1167.00	Elite
14	535.00	Elite
15	246.50	Good
16	124.00	Good
17	1047.50	Elite
18	1235.00	Elite

---

7. Categorize Constructors by Region Based on Nationality

Objective: Group constructors into European or Non-European regions.

Hints:

- Identify which nationalities belong to each region.
- Assign each constructor a region based on their nationality.
- Present a clear summary of each constructor's region.
- **Expected :**

constructorId	name	region
1	McLaren	European
2	BMW Sauber	European
3	Williams	European
4	Renault	European
5	Toro Rosso	European
6	Ferrari	European
7	Toyota	Non-European
8	Super Aguri	Non-European
9	Red Bull	Non-European
10	Force India	Non-European
11	Honda	Non-European
12	Spyker	European
13	MF1	Non-European
14	Spyker MF1	European
15	Sauber	Non-European
16	BAR	European
17	Jordan	Non-European
18	Minardi	European

---

8. Identify Drivers Who Never Raced for Ferrari

Objective: Find drivers who participated in races but never drove for a specific constructor.

Hints:

- Determine all drivers who participated in any race.
- Exclude drivers associated with the specific constructor.
- Return the names of drivers meeting these criteria.
- **Expected :**

forename	surname
Carlo	Abate
George	Abecassis
Kenny	Acheson
Philippe	Adams
Walt	Ader
Fred	Agabashian
Kurt	Ahrens
Jack	Aitken
Christijan	Albers
Alexander	Albon
Jaime	Alguersuari
Philippe	Alliot
Giovanna	Amati
George	Amick
Red	Amick
Bob	Anderson
Conny	Andersson
Michael	Andretti

---

9. List Races Held on Circuits in the UK

Objective: Identify races hosted at UK circuits.

Hints:

- Find circuits located in the UK.
- Use this information to filter the races held there.
- Include race names in your results.
- **Expected :**

name
British Grand Prix
70th Anniversary Grand Prix
European Grand Prix

---

10. Calculate Average Points Per Race for Top Drivers

Objective: Determine which drivers perform best on average per race.

Hints:

- Compute the total points and the number of races per driver.
- Consider only drivers with substantial participation.
- Calculate average points and rank drivers accordingly.
- **Expected :**

forename	surname	average_points_per_race	total_races
Max	Verstappen	13.935407	209
Lewis	Hamilton	13.540730	356
Sebastian	Vettel	10.326667	300
Charles	Leclerc	9.147651	149
Nico	Rosberg	7.740291	206
Lando	Norris	7.421875	128
Valtteri	Bottas	7.238866	247
Carlos	Sainz	5.786058	208
Fernando	Alonso	5.764851	404
Sergio	Pérez	5.600707	283
Kimi	Räikkönen	5.321023	352
George	Russell	5.187500	128
Daniel	Ricciardo	5.136187	257
Michael	Schumacher	5.084416	308
Mark	Webber	4.827189	217
Felipe	Massa	4.306273	271
Jenson	Button	3.996764	309
Alain	Prost	3.952970	202

---

11. Find the Most Common Race Status

Objective: Identify the top 5 most frequent race outcomes.

Hints:

- Aggregate race results by status.
- Determine the frequency of each status.
- Focus on the statuses that occur most often.
- **Expected :**

status	status_count
Finished	7674
+1 Lap	4037
Engine	2026
+2 Laps	1613
Accident	1062

---

12. Calculate Average Lap Times for Drivers Who Won at Least One Race

Objective: Determine how winning drivers perform on average per lap.

Hints:

- Identify drivers who have won at least one race.
- Calculate the average lap time for these drivers.
- Group your results per driver.
- **Expected :**

driverId	avg_lap_time
1	96752.7083
3	98486.7227
4	96208.6093
8	93918.7793
9	95376.1733
13	96249.2884
15	93661.5261
17	95273.1087
18	95662.1676
20	97876.8814
21	90985.8294
22	92768.2231
5	100620.9578
14	89663.8107

---

13. Identify Constructors' First Win Year

Objective: Find the first year each constructor won a race.

Hints:

- Combine constructor information with race results and years.
- Focus on the earliest successful race per constructor.
- Present a summary showing each constructor and their first win year.
- **Expected :**

constructor_name	first_win_year
Alfa Romeo	1950
Kurtis Kraft	1950
Ferrari	1951
Kuzma	1952
Maserati	1953
Mercedes	1954
Watson	1956
Epperly	1957
Vanwall	1957
Cooper	1958
BRM	1959
Cooper-Climax	1959
Team Lotus	1960
Lotus-Climax	1961

---

14. Identify Peak Performance Seasons

Objective: Determine the seasons with the highest points scored by any constructor.

Hints:

- Calculate total points for each constructor per season.
- Determine the maximum points achieved in each season.
- Identify the top-performing seasons.
- **Expected :**

season	max_points
2023	790.00
2016	765.00
2019	739.00
2022	724.00
2015	703.00